Report: Full

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Game level

- Zone% was 49% vs 48%, Whiff% 17% vs 18%, Chase% 26% vs 22%, and CSW% 26% vs 29%. The pitching and swing rates hardly changed game to game.
- Strikeouts were nearly the same (19 vs 18) and walks were close (5 vs 6). The difference in results came from hits (23 vs 10) and home runs (2 vs 0), which rates alone don't explain.
- Average velocity was steady, 89.8 mph in Game 1 and 90.3 mph in Game 2.

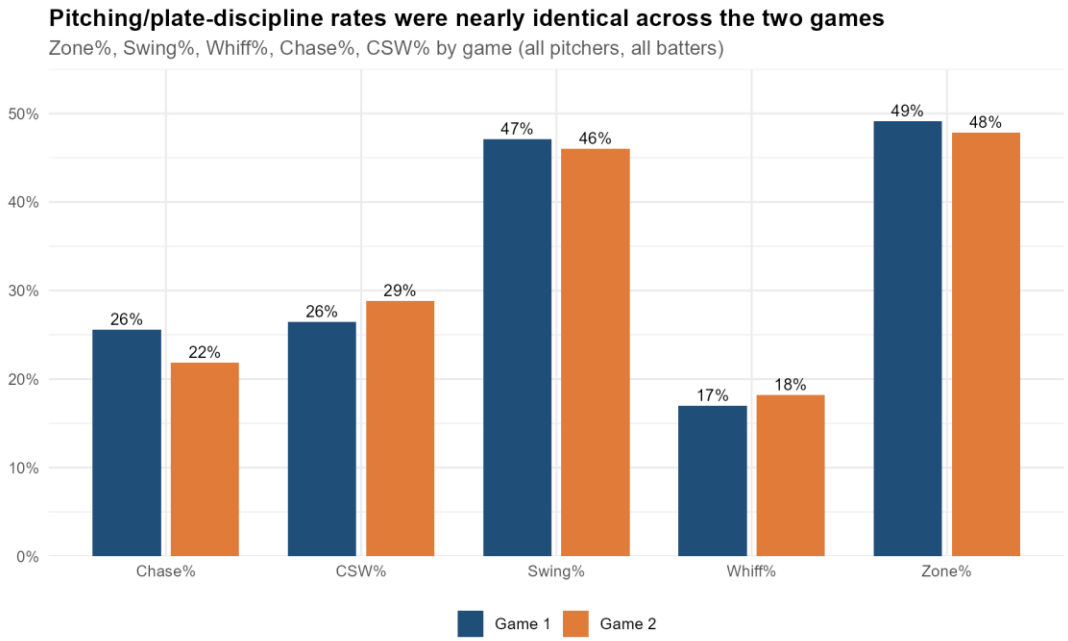


Figure 1. Zone%, Swing%, Whiff%, Chase% and CSW% by game (all pitchers, all batters).

Table 1. Game summary (all pitchers and batters).

Game	Pitches	PA	H	HR	K	BB	Zone%	Swing%	Whiff%	Chase%	CSW%	Avg Velo
Game 1	344	86	23	2	19	5	49%	47%	17%	26%	26%	89.8
Game 2	257	73	10	0	18	6	48%	46%	18%	22%	29%	90.3

Pitchers

- Starters threw 75-105 pitches (P1, P11 and P7), and relievers threw 6-29, so only starters have enough data to judge individually.

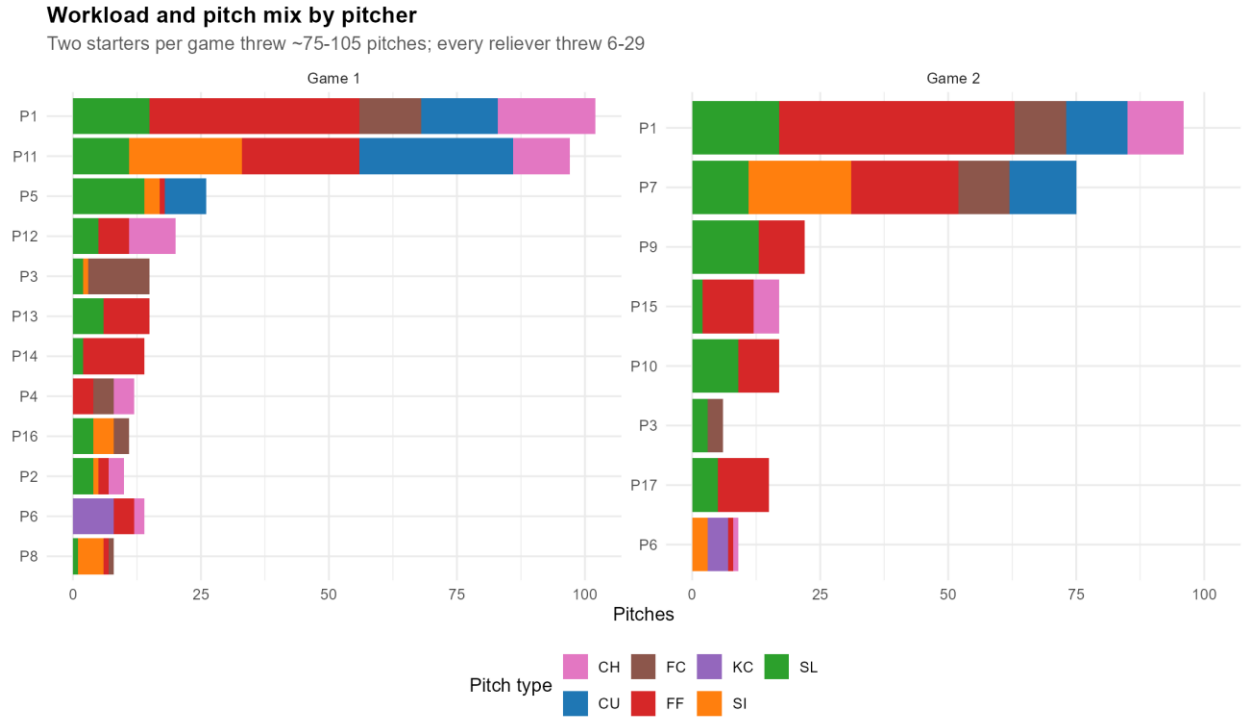


Figure 2. Pitches thrown and pitch mix by pitcher, by game.

Table 2. Pitcher performance, both games combined.

Pitcher	Pitches	PA	H	HR	K	BB	Zone%	Swing%	Whiff%	Chase%	CSW%	Avg Velo
P1	198	47	8	1	13	3	45%	50%	17%	31%	28%	89.5
P2	10	3	0	0	2	0	50%	50%	0%	25%	40%	92.4
P3	21	5	1	0	3	0	48%	39%	29%	10%	33%	92.0
P4	12	4	0	0	0	1	50%	33%	0%	0%	17%	89.1
P5	26	5	2	0	0	0	65%	50%	23%	33%	23%	88.5
P6	23	6	0	0	2	0	61%	29%	0%	22%	43%	92.0
P7	75	26	4	0	5	2	44%	51%	19%	29%	27%	89.3
P8	8	3	0	0	0	1	25%	25%	0%	17%	0%	96.6
P9	22	4	1	0	3	0	64%	42%	12%	25%	36%	90.7
P10	17	6	2	0	0	1	47%	35%	17%	0%	24%	90.9
P11	97	26	8	0	5	1	47%	50%	15%	29%	26%	88.5
P12	20	6	3	0	1	0	55%	53%	30%	22%	35%	93.2
P13	15	3	1	0	0	0	60%	47%	29%	0%	20%	91.7
P14	14	5	1	1	0	1	57%	43%	0%	0%	7%	91.5
P15	17	3	0	0	2	0	53%	47%	29%	12%	29%	92.9
P16	11	2	1	0	1	0	36%	40%	50%	17%	36%	88.4
P17	15	5	1	0	0	1	40%	33%	0%	0%	20%	92.1

- P1 threw 198 pitches across both games, with 13 strikeouts and 3 walks against 47 batters.
- P1's mix was similar in both games, with four-seamers at 40% and 48%. His results differed between six hits including a home run in Game 1 and two hits in Game 2.

Table 3. P1 performance by game.

Game	Pitcher	Pitches	PA	H	HR	K	BB	Zone%	Swing%	Whiff%	Chase%	CSW%	Avg Velo
Game 1	P1	102	22	6	1	6	1	43%	50%	17%	36%	28%	89.1
Game 2	P1	96	25	2	0	7	2	48%	49%	18%	24%	28%	89.9

- Pitch mix differs by pitcher: P1 leans on four-seamers, P11 relies on the curveball (31%), and P7 split sinkers (27%) and four-seamers (28%).

Table 4. Starters' pitch mix (share of pitches).

Game	Pitcher	CH	CU	FC	FF	SL	SI
Game 1	P1	19%	15%	12%	40%	15%	0%
Game 1	P11	11%	31%	0%	24%	11%	23%
Game 2	P1	11%	12%	10%	48%	18%	0%
Game 2	P7	0%	17%	13%	28%	15%	27%

- Starters' four-seam velocity stayed at 93-96 mph into the late innings.

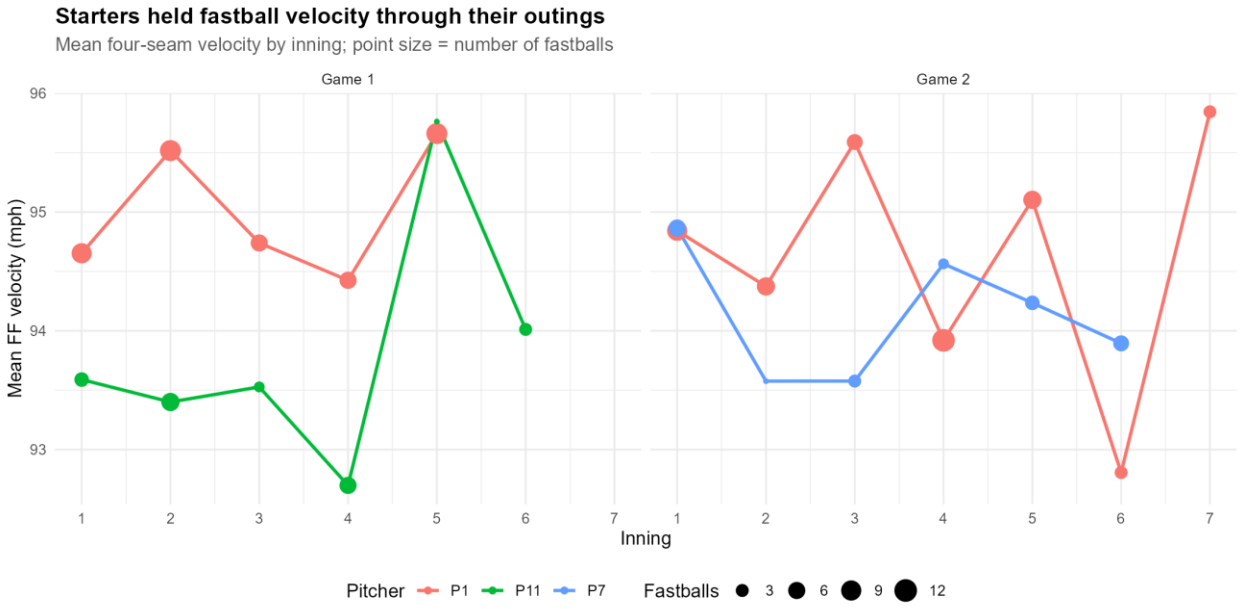


Figure 3. Mean four-seam velocity by inning for the starters; point size = number of fastballs.

Pitch types

- Sliders (124 pitches) had a 36% whiff rate and 37% CSW, the best of any pitch type.
- Four-seamers (208 pitches) had a 10% whiff rate and 22% CSW. Sinkers (59 pitches) were similar, at 10% and 20%.
    - Pitches thrown most were among the least effective at missing bats.

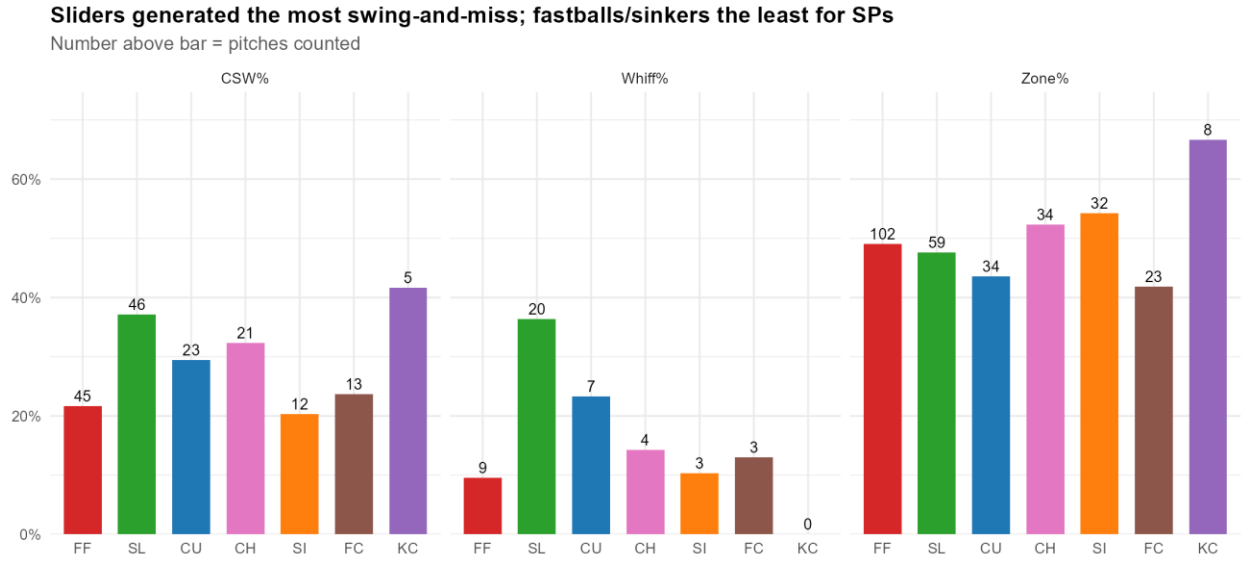


Figure 4. Whiff%, CSW% and Zone% by pitch type. The number above each bar is the count of pitches behind the rate (whiffs, called + swinging strikes, or in-zone pitches).

Table 5. Pitch-type summary (all pitchers).

Pitch	Pitches	Usage%	Zone%	Whiff%	Chase%	CSW%	Avg Velo	Avg spin	Horiz (in)	Vert (in)
FF	208	35%	49%	10%	23%	22%	95.0	2380	4.4	14.0
SL	124	21%	48%	36%	29%	37%	86.9	2352	-3.8	1.9
CU	78	13%	44%	23%	26%	29%	80.3	2584	-6.5	-8.0
CH	65	11%	52%	14%	17%	32%	87.3	1607	7.4	3.5
SI	59	10%	54%	10%	22%	20%	94.4	2318	1.3	6.9
FC	55	9%	42%	13%	23%	24%	91.2	2458	-1.7	8.3
KC	12	2%	67%	0%	25%	42%	88.5	2862	-5.1	-9.2

- Curveballs were average (23% whiff, 29% CSW).
- Change-ups missed less (14% whiff) but had a 32% CSW.
- The knuckle-curve was thrown only 12 times.
- Fastballs and sinkers had the tightest velocity ranges.
- P7's slider profiles like a sweeper; about 17 inches of horizontal break, little vertical movement, and marginally slower.

See Appendix Figure A1 for velocity by pitch type.

Count and release

- P1 threw a four-seamer on 24 of 40 pitches when behind in the count, and used change-ups on 18 of 71 pitches with two strikes. His CSW% at two strikes was only 18%, his lowest of any count group.
- The count results for P7 and P11 rest on about 4-28 pitches per group.
    - Too little to make claims from

Table 6. Starters' pitch usage and CSW% by count, both games combined.

Pitcher	Count	Pitches	Four-seamers	Change-ups	CSW%
P1	First pitch	47	19	2	28%
P1	Ahead	23	9	3	39%
P1	Even	17	11	4	41%
P1	Behind	40	24	3	35%
P1	Two strikes	71	24	18	18%
P7	First pitch	26	8	0	31%
P7	Ahead	11	1	0	18%
P7	Even	4	0	0	50%
P7	Behind	22	10	0	14%
P7	Two strikes	12	2	0	42%
P11	First pitch	26	7	3	27%
P11	Ahead	10	5	1	20%
P11	Even	11	2	0	18%
P11	Behind	28	5	7	32%
P11	Two strikes	22	4	0	23%

Count is the count before the pitch: First pitch (0-0), Ahead (more strikes than balls, under two strikes), Even, Behind (more balls than strikes), Two strikes. Computed from the pitch-level file with the same definitions as the other tables.

- Starters' release points were tight, mostly within 1-2 inches for each pitch type. P11's slider was the loosest, at about 3 inches horizontally.
- P1's curveball was released about 2 inches higher than his change-up
    - Could affect his deception.

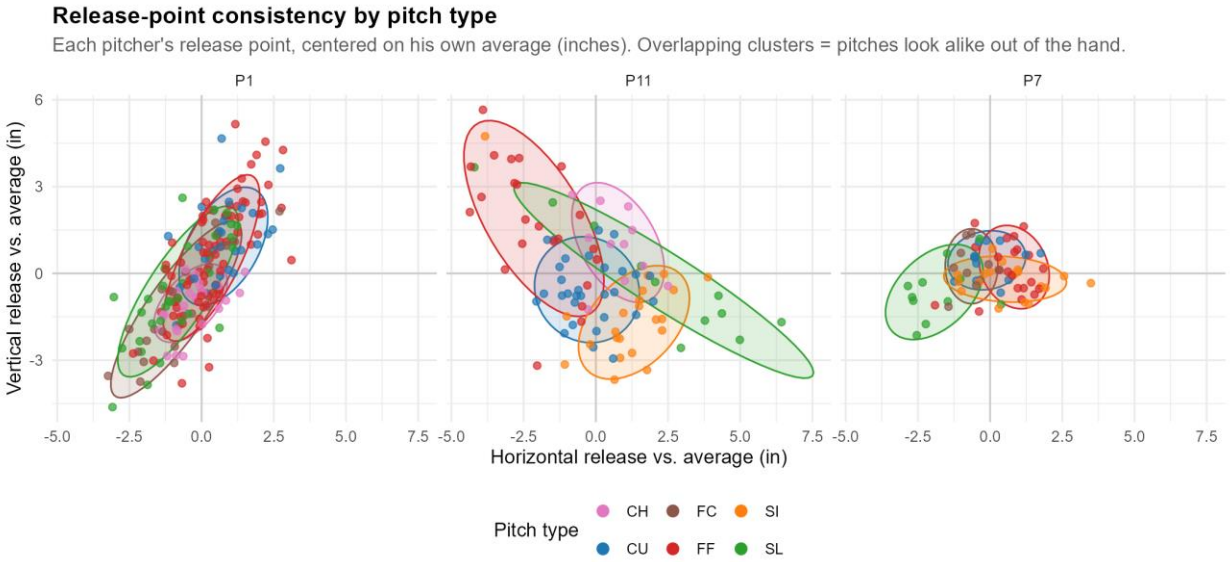


Figure 5. Release-point consistency by pitch type for the starters. Each pitcher's release point is centered on his own average (inches); overlapping ellipses mean a consistent release point.

Table 7. Release-point spread by pitcher and pitch type (standard deviation, inches).

Pitcher	Pitch	Pitches	Avg extension (ft)	Horiz SD (in)	Vert SD (in)
P1	FF	87	6.14	1.1	1.8
P1	SL	32	6.08	1.3	1.8
P1	CH	30	6.31	0.8	0.9
P1	CU	27	6.30	1.0	1.4
P1	FC	22	6.10	1.4	1.8
P7	FF	21	6.49	0.9	0.9
P7	SI	20	6.54	1.3	0.5
P7	CU	13	6.38	0.8	0.6
P7	SL	11	6.42	1.1	1.0
P7	FC	10	6.43	0.5	0.8
P11	CU	30	6.35	1.1	1.1
P11	FF	23	6.53	1.4	2.1
P11	SI	22	6.40	1.6	1.7
P11	CH	11	6.31	1.0	1.3
P11	SL	11	6.44	3.1	2.0

Batters

- B21 and B24 had the most hits, 5 and 4 in 10 plate appearances each. B21 hit .556.
- B19 struck out in 4 of 10 plate appearances, and also had a 36% whiff rate.

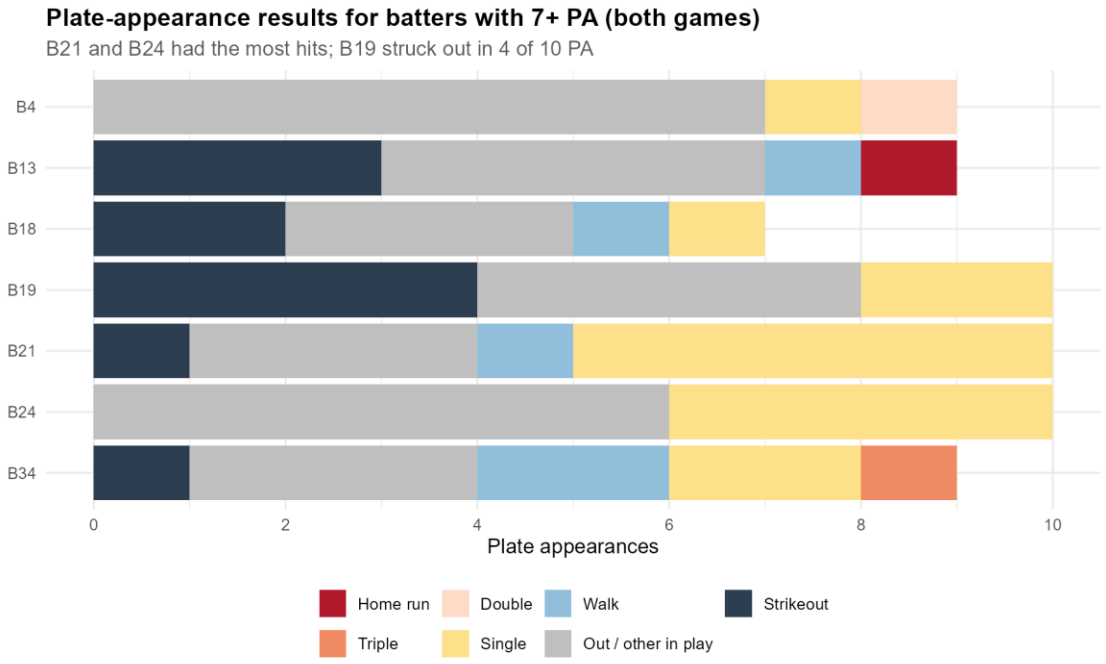


Figure 6. Plate-appearance results for batters with 7+ plate appearances (both games).

- B4 chased outside the zone on 43% of swings, and B14 on 73%, though B14's rate comes from only 17 pitches.

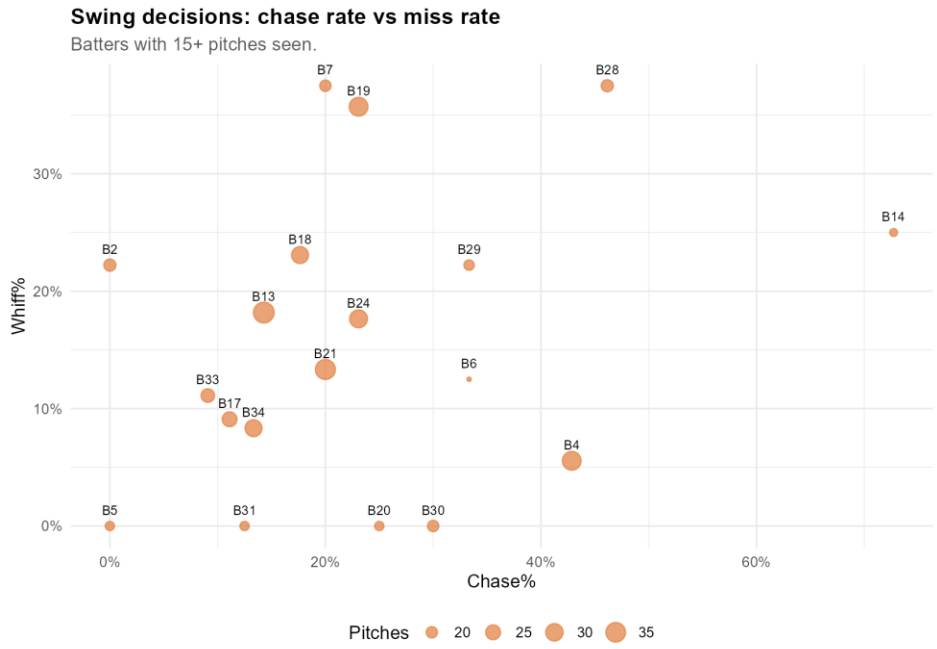


Figure 7. Chase% vs. Whiff% for batters who saw 15+ pitches; point size = pitches seen.

- B13 and B20 each hit a home run.

Table 8. Batters named in the conclusions (full batter table in the Appendix).

Batter	PA	AB	H	HR	K	BB	AVG	Pitches	Swing%	Whiff%	Chase%	Zone%
B4	9	9	2	0	0	0	.222	33	55%	6%	43%	58%
B13	9	8	1	1	3	1	.125	38	31%	18%	14%	42%
B14	5	5	2	0	1	0	.400	17	75%	25%	73%	29%
B19	10	10	2	0	4	0	.200	33	48%	36%	23%	58%
B20	5	5	1	1	0	0	.200	18	50%	0%	25%	56%
B21	10	9	5	0	1	1	.556	36	43%	13%	20%	44%
B24	10	10	4	0	0	0	.400	31	55%	18%	23%	58%

- Left-handed pitchers struck out no left-handed batters in 18 plate appearances, while those batters had 8 hits.
- Right-handed pitchers struck out right-handed batters 20 times in 65 plate appearances.

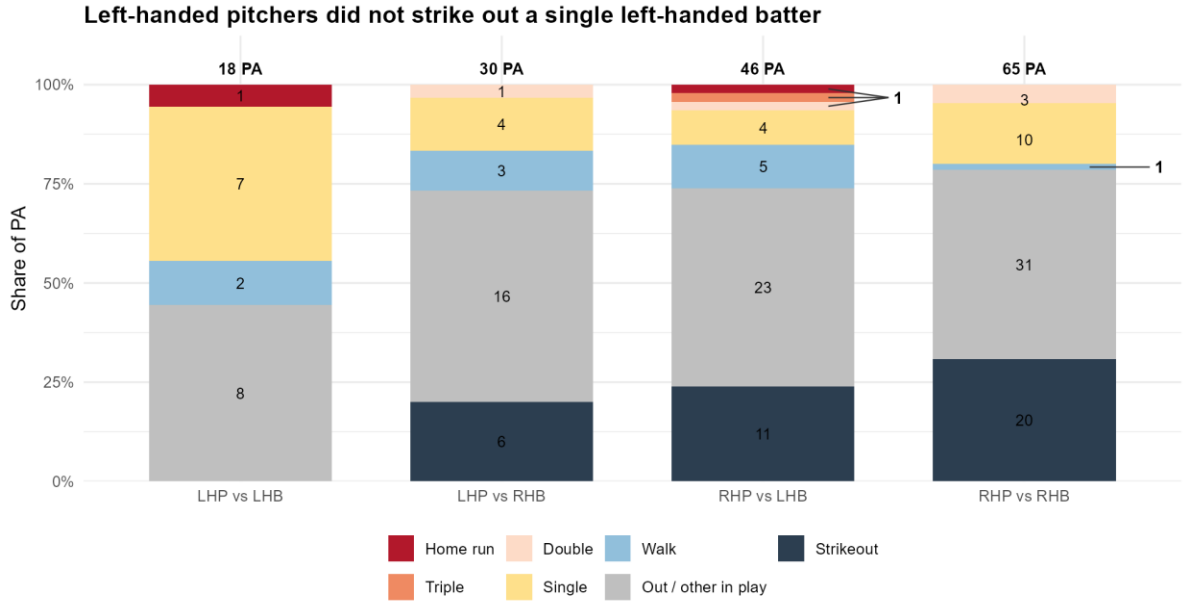


Figure 8. Plate-appearance results by matchup (LHP/RHP = pitcher hand, LHB/RHB = batter side). Numbers inside bars are counts; total plate appearances are above each bar.

Caveats

- Most players have fewer than 25 pitches or 10 plate appearances.
- Strikeouts are left out of swing-based rates because the data doesn't say swinging or looking.

Appendix

Additional figures and reference tables supporting the conclusions above.

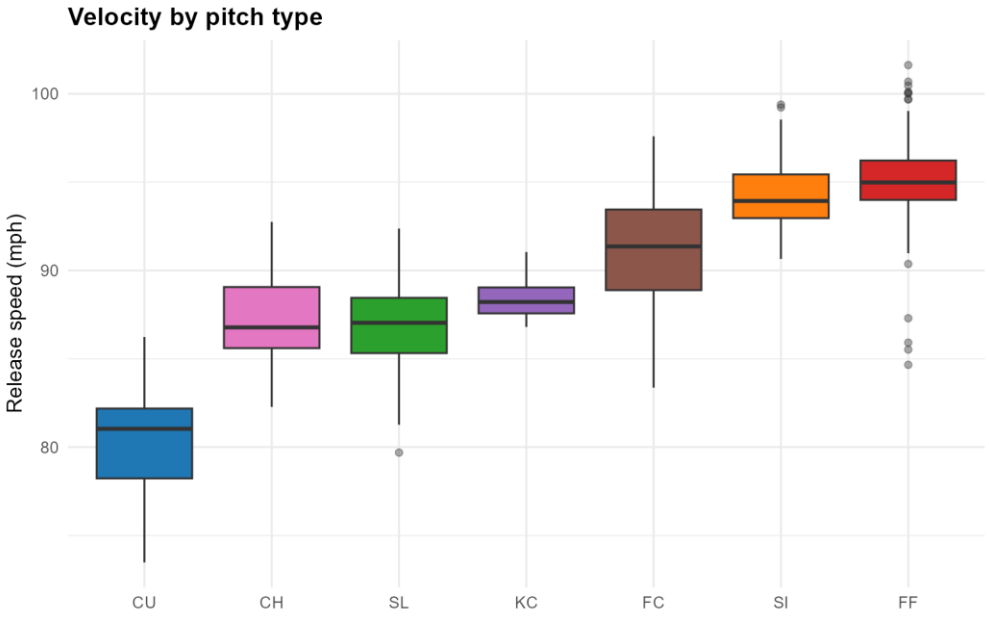


Figure A1. Release velocity by pitch type (all pitchers, both games).

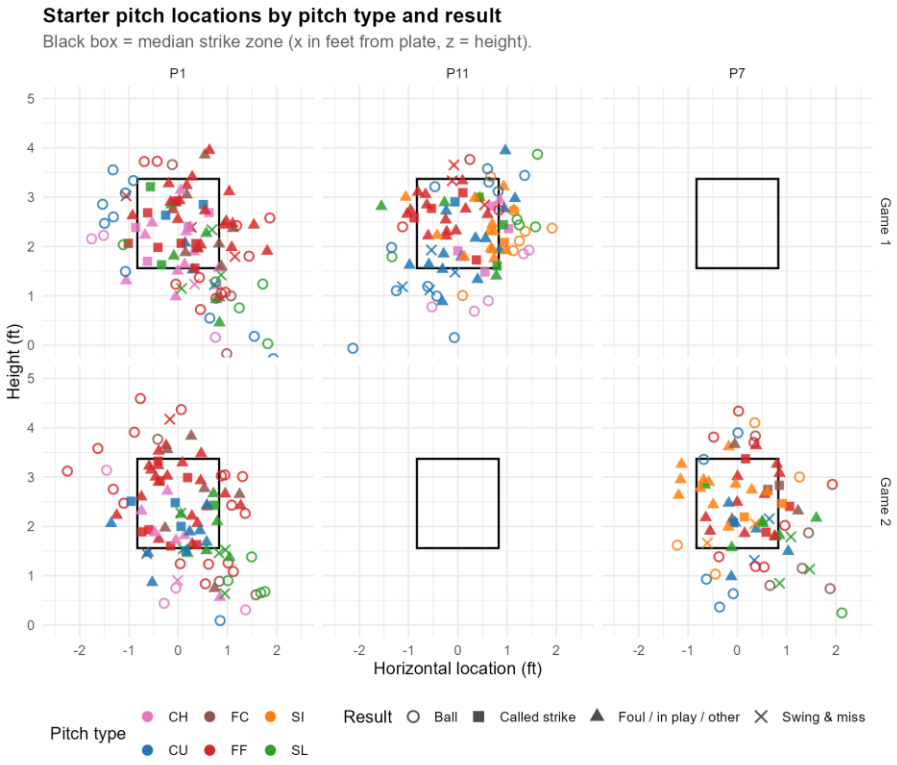


Figure A2. Pitch locations for the starters. Color = pitch type, shape = result. The black box is the median strike zone (x in feet from plate center, z = height).

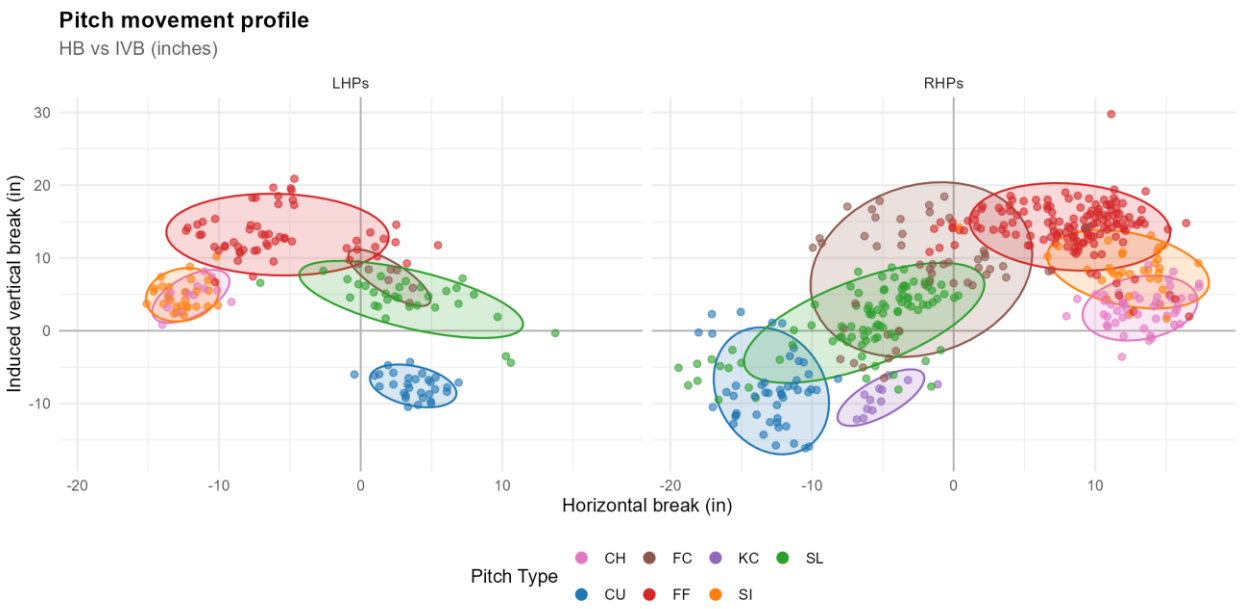


Figure A3. Pitch movement profile by pitcher hand: horizontal break vs. induced vertical break (inches). Ellipses enclose about 80% of each pitch type.

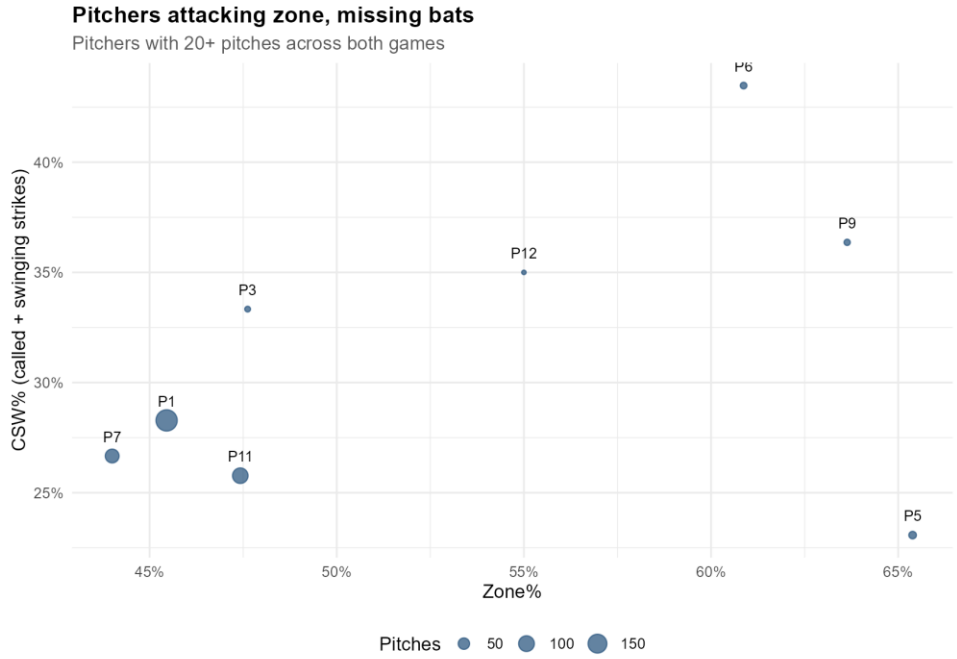


Figure A4. Zone% vs. CSW% for pitchers with 20+ pitches across both games; point size = pitches thrown.

Table A1. All batters, both games combined.

Batter	PA	AB	H	HR	K	BB	AVG	Pitches	Swing%	Whiff%	Chase%	Zone%
B1	2	2	1	0	0	0	.500	5	60%	33%	100%	80%
B2	5	5	0	0	4	0	.000	21	53%	22%	0%	62%
B3	1	1	1	0	0	0	1.000	2	50%	0%	-	100%
B4	9	9	2	0	0	0	.222	33	55%	6%	43%	58%
B5	4	1	0	0	0	3	.000	18	17%	0%	0%	28%
B6	4	4	1	0	1	0	.250	16	53%	12%	33%	38%
B7	5	5	2	0	1	0	.400	20	42%	38%	20%	50%
B8	2	2	0	0	1	0	.000	9	62%	20%	67%	56%
B9	1	1	0	0	1	0	.000	6	40%	50%	0%	50%
B10	1	1	0	0	1	0	.000	6	20%	100%	25%	33%
B11	3	3	0	0	2	0	.000	11	44%	50%	25%	45%
B12	4	4	1	0	1	0	.250	13	58%	29%	17%	46%
B13	9	8	1	1	3	1	.125	38	31%	18%	14%	42%
B14	5	5	2	0	1	0	.400	17	75%	25%	73%	29%
B15	4	4	0	0	1	0	.000	11	60%	33%	0%	64%
B16	3	3	0	0	0	0	.000	7	43%	0%	0%	71%
B17	5	4	1	0	1	1	.250	25	46%	9%	11%	60%
B18	7	6	1	0	2	1	.167	29	48%	23%	18%	41%
B19	10	10	2	0	4	0	.200	33	48%	36%	23%	58%
B20	5	5	1	1	0	0	.200	18	50%	0%	25%	56%
B21	10	9	5	0	1	1	.556	36	43%	13%	20%	44%
B22	1	1	0	0	1	0	.000	5	75%	0%	50%	40%
B23	2	1	1	0	0	1	1.000	10	50%	40%	29%	30%
B24	10	10	4	0	0	0	.400	31	55%	18%	23%	58%
B25	3	3	0	0	0	0	.000	12	67%	0%	57%	42%
B26	5	4	0	0	1	0	.000	11	60%	17%	25%	64%
B27	3	3	0	0	0	0	.000	7	57%	0%	50%	43%
B28	4	4	0	0	3	0	.000	21	44%	38%	46%	24%
B29	5	5	0	0	2	0	.000	19	53%	22%	33%	47%
B30	4	4	3	0	1	0	.750	20	37%	0%	30%	50%
B31	4	4	1	0	1	0	.250	18	35%	0%	12%	56%
B32	1	1	0	0	0	0	.000	2	50%	0%	0%	50%
B33	5	5	0	0	2	0	.000	23	43%	11%	9%	43%
B34	9	7	3	0	1	2	.429	29	43%	8%	13%	48%
B35	1	0	0	0	0	1	-	6	17%	0%	0%	33%
B36	2	2	0	0	0	0	.000	10	30%	0%	0%	60%
B37	1	1	0	0	0	0	.000	3	33%	0%	0%	67%

Table A2. Pitcher glossary: pitcher IDs and the labels used in this report.

Pitcher ID	Label used	Hand	Games	Pitches
1	P1	R	1, 2	198
2	P2	R	1	10
3	P3	R	1, 2	21
4	P4	L	1	12
5	P5	R	1	26
6	P6	R	1, 2	23
7	P7	R	2	75
8	P8	R	1	8
9	P9	R	2	22
10	P10	L	2	17
11	P11	L	1	97
12	P12	R	1	20
13	P13	R	1	15
14	P14	L	1	14
15	P15	R	2	17
16	P16	L	1	11
17	P17	L	2	15

Definitions and data notes

- Zone% is the share of pitches in the strike zone (within 0.83 ft of plate center horizontally and within the batter's zone top/bottom, with about a ball radius of margin).
- Whiff% is swinging strikes divided by swings. Chase% is the share of swings at pitches outside the zone. CSW% is called plus swinging strikes (including strike three) divided by all pitches.
- Strikeout pitches are left out of swing-based rates because the data does not say whether strike three was swinging or looking.
- Batting average is hits divided by at-bats (plate appearances minus walks and sacrifice bunts).
- Data: 601 pitches across two games. Pickoff throws were excluded. P# and B# labels come from the pitcher and batter IDs in the data (pitcher and batter IDs are separate).